

6.3 #5: 1, 2

6.4 #5: 2, 3, 4, 10

6.5 #5: 2, 4, 7, 11

Homework 11, Math 3210

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Determine if the following series converge absolutely, converge conditionally or diverge. Justify your answer.

1. $\sum_{k=1}^{\infty} \frac{(-1)^k}{k^{1/3}}$: By Example 6.3.3, we learn that alternating p -series with $0 < p \leq 1$ converges conditionally. We have exactly that here where $p = 1/3$. However, $\sum_{k=1}^{\infty} \left| \frac{(-1)^k}{k^{1/3}} \right|$ does not converge because we have a p -series with $p = 1/3 < 1$. So, $\sum_{k=1}^{\infty} \frac{(-1)^k}{k^{1/3}}$ converges conditionally. ■

2. $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2}$: From 6.1.10, we have $\sum |a_k|$ converges implies $\sum a_k$ converges absolutely. In our case here, we have $\sum \left| \frac{(-1)^{k+1}}{k^2} \right| = \sum \frac{1}{k^2}$ which converges by $a_{k=1}$ p -series with $p = 2 > 1$. Thus, by 6.1.10 or 6.1.11, $\sum \frac{(-1)^{k+1}}{k^2}$ converges absolutely. ■

#2: Prove the function $f(x) = \sum_{k=1}^{\infty} \frac{\sin kx}{2^k}$ is continuous on the entire real line.

We will prove this using W-M-Test, Ex. 6.4.5 and Theorem 6.4.2. $\left| \frac{\sin kx}{2^k} \right| \leq \frac{1}{2^k}$. The series $\sum_{k=1}^{\infty} \frac{1}{2^k} = \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k$ converges by Ratio Test where $r = \frac{\left(\frac{1}{2}\right)^{k+1}}{\left(\frac{1}{2}\right)^k} = \frac{1}{2} < 1$, thus converges absolutely. ■

Theorem 6.4.2 implies $f(x)$ is cont. on $\mathbb{R}$. ■

#4: $\sum_{k=1}^{\infty} \left(\frac{1}{k^3}\right)^{x/k}$: $R = \frac{1}{\limsup_{k \rightarrow \infty} \left(\frac{1}{k^3}\right)^{1/k}}$: $\limsup \left| \frac{1}{k^3} \right|^{1/k} = \lim \left(\frac{1}{3}, \sqrt{\frac{1}{2}}, \sqrt[3]{\frac{1}{3}}, \dots \right) \rightarrow \frac{1}{3}$: $\left\{ \left(\frac{1}{k}\right)^{1/k} \right\} \rightarrow 1$
 $\frac{1}{1/3} = 3$. So, $R = 3$. ■

6.4 #5: 3, 4, 10 — Assume each f_k' is continuous.

6.5 #5: 2, 4, 7, 11

Homework 11, cont.

6.4 #3: Let $\{f_k\}$ be a sequence of differentiable functions on (a, b) and suppose there is a point $c \in (a, b)$ such that the series $\sum_{k=1}^{\infty} f_k(c)$ converges. Suppose also that the seq. of derivatives $\{f_k'\}$ satisfies $|f_k'(x)| \leq M_k$ on (a, b) and the series $\sum_{k=1}^{\infty} M_k$ converges. Then prove that the series defining

$f(x) = \sum_{k=1}^{\infty} f_k(x)$ and $g(x) = \sum_{k=1}^{\infty} f_k'(x)$ converge on (a, b) and f is differentiable with derivative g on (a, b) .

Claim 1: $f(x) = \sum_{k=1}^{\infty} f_k(x)$, $g(x) = \sum_{k=1}^{\infty} f_k'(x)$ converge uniformly on (a, b) .

Proof: Since $|f_k'(x)| \leq M_k \forall x \in (a, b)$ and $\sum_{k=1}^{\infty} M_k$ converges, $g(x)$ converges uniformly by the W-M-Test.

Consider $\{S_n(x)\}$ for $\sum_{k=1}^n f_k(x)$. This converges for some $c \in (a, b)$. Since each derivative is bounded by some M_k , it follows that $f(x)$ is uniformly continuous.

Claim 2: f is differentiable with derivative g on (a, b) .

$$f'(x) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \left[\frac{\lim_{n \rightarrow \infty} \sum_{k=1}^n f_k(x) - f(a)}{x - a} \right]$$

$$= \lim_{x \rightarrow a} \left[\frac{\lim_{n \rightarrow \infty} f_1(x) + f_2(x) + f_3(x) + \dots + f_n(x) - f(a)}{x - a} \right]$$

Since each $f_k(x)$ is continuous, (differentiable), we get:

$$\lim_{n \rightarrow \infty} \frac{f_1(a) + f_2(a) + f_3(a) + \dots + f_n(a) - f(a)}{x - a} \rightarrow \lim_{n \rightarrow \infty} \sum_{k=1}^n f_k'(x) = g(x).$$

6.4 #5: 10

6.5 #5: 2, 4, 7, 11

Homework 11, cont.

- 10: Let $\{a_k\}$ be a non-increasing sequence of non-negative numbers which converges to zero. Use Theorem 6.3.2 to show that the power series $\sum_{k=0}^{\infty} (-1)^{k+1} a_k x^k$ converges uniformly on $[0, 1]$ and, hence, converges to a continuous function on this interval.

Def. 6.4.1: $\sum_{k=1}^{\infty} f_k(x)$ converges uniformly on $I \subset \mathbb{R}$ if $\{S_n(x)\}$ converges uniformly on I where $S_n(x) = \sum_{k=1}^n f_k(x)$.

Theorem 6.3.2. Alt. Series Test: Let $\{a_k\}$ be of the above form. Then $\sum_{k=1}^{\infty} (-1)^{k+1} a_k = a_1 - a_2 + a_3 - a_4 + \dots$ converges. In fact, if S_n is the n^{th} partial sum of this series and $S = \lim S_n$, then $|S - S_n| \leq a_{n+1}$ for all n .

We have $\sum_{k=0}^{\infty} (-1)^{k+1} a_k x^k = a_1 x - a_2 x^2 + a_3 x^3 - a_4 x^4 + \dots$

converges to some function over an interval.

Likewise, by Def. 6.4.1 a series of functions converges uniformly on $[0, 1]$ if $\{S_n(x)\}$ converges uniformly on $[0, 1]$.

Using the final ~~line~~ and definition of uniform convergence, we show that $\sum_{k=1}^{\infty} (-1)^{k+1} a_k x^k = a_1 x - a_2 x^2 + \dots + (-1)^{n+1} a_n x^n$.

This series converges by AST and converges uniformly by final line of AST on $[0, 1]$.

By Theorem 6.4.2, the function to which it converges is also continuous on $[0, 1]$. $\square$

6.5 #5: 2, 4, 7, 11

Homework 11, cont.

6.5 #2: Find the Taylor Series expansion for $\cos x$ at 0 and show that it converges for all x .

$$f(x) = \cos x = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} (x)^k$$

$$f(0) = 1, \quad f'(0) = -\sin(0) = 0, \quad f''(0) = -\cos(0) = -1$$

$$f^{(3)}(0) = \sin(0) = 0$$

$$f^{(k)}(0) = (-1)^n; \quad \cos x = \sum_{k=0}^{\infty} \frac{1}{(k+1)!} x^k + \sum_{k=2}^{\infty} \frac{-1}{(k+4)!} x^k$$

$$= \sum_{k=0}^{\infty} \frac{1 - (k+4)(k+3)}{(k+4)!} x^k$$

$$R = \frac{1}{\limsup \left| \frac{f^{(k)}(0)}{k!} \right|^{1/k}} = 0$$

So, $R = \infty$ by definition. Thus, converges for all x .

$$4: e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \Rightarrow e = \sum_{k=0}^{\infty} \frac{1}{k!} = \sum_{k=0}^n \frac{1}{k!} + R_n(1)$$

$$R_n(1) = \frac{f^{(n+1)}(1)}{(n+1)!} (1)^{n+1} \in \mathbb{R}$$

$$\text{Want: } R_n(1) < 0.001$$

$$\text{Know that } \frac{f^{(n+1)}(1)}{(n+1)!} = \frac{e}{(n+1)!} < \frac{e}{(n+1)!} < \frac{3}{(n+1)!}$$

$$\text{Solving } \frac{e}{(n+1)!} < 0.001 \text{ guarantees } R_n(1) < 0.001.$$

$$e < 0.001 (n+1)! \Rightarrow [n=6]$$

$$7: f(x) = \ln(1+x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} (x)^k = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} (k-1)!}{k!} x^k$$

$$k=0, \ln(1+x); \ln(1)=0$$

$$k=1, \frac{1}{1+x} = \frac{1}{1+0} = 1$$

$$k=2, \frac{-1}{(1+x)^2} = \frac{-1}{1} = -1$$

$$k=3, \frac{2}{(1+x)^3} = 2$$

$$R = \frac{1}{\limsup \left| \frac{f^{(k)}(0)}{k!} \right|^{1/k}} = \limsup \left(\frac{1}{k} \right)^{1/k} \rightarrow 1$$

(Previously encountered $\limsup \left(\frac{1}{k} \right)^{1/k}$)

$$\text{So, } R=1$$

6.5 Hs: 11

Homework 11, cont.

- 11 Prove that if f is an infinitely diff. function on $(a-r, a+r)$ and there is a constant k such that $|f^{(n)}(x)| \leq k \frac{n!}{r^n}$ for all $n \in \mathbb{N}$ and all $x \in (a-r, a+r)$, then the Taylor Series for f at a converges to f on $(a-r, a+r)$.

$$\sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$$

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k \text{ converges to } f(x) \text{ if } \lim_{n \rightarrow \infty} |R_n(x)| = 0.$$

We have a formula for $R_n(x)$:

$$|R_n(x)| = \left| \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1} \right| \quad ; \quad c \in [a, x]$$

$$\leq \frac{M \frac{(n+1)!}{r^{n+1}}}{(n+1)!} |x-a|^{n+1} = \frac{M}{r^{n+1}} |x-a|^{n+1}$$

$$\text{note: } \begin{cases} a-r < x < a+r \\ -r < x-a < r \\ -1 < \frac{x-a}{r} < 1 \end{cases}$$

$$= M \left(\frac{|x-a|}{r} \right)^{n+1}$$

By note, it follows that $-1 < \frac{|x-a|}{r} < 1$. This implies $M \left(\frac{|x-a|}{r} \right)^{n+1} \rightarrow 0$ for any constant M .

Thus, $|R_n(x)| \rightarrow 0$ by Theorem $|a_n| \leq |b_n|$ and $|b_n| \rightarrow 0$, then $|a_n| \rightarrow 0$.